Polygon-based unified Fourier-modal approach for diffractive-optics simulations

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In this article, we derive a theoretical formalism that unifies the rigorous coupled wave analysis and the dynamical diffraction theory. Based on this formalism, we design a computational approach for the diffraction calculation for the nanoscale lamellar gratings with an arbitrary line profile shape. In this approach, the gratings line profile is approximated as a polygon. This proves to be convenient since such an approach does not rely on the geometry model of the grating. We test the proposed approach against other computational theories and a synchrotron scattering experiment.

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I. INTRODUCTION

The problem of monochromatic wave scattering on a periodic potential arises in many fields of science and technology. Starting with early x-ray diffraction on crystals research, it is now in photonics [1] and plasmonics [2], engineering of astronomical equipment [3], and diffractive optics, which, for instance, now have applications in augmented reality devices [4]. The synchrotron nanometrology is yet another research field in which the scattering on periodic systems is employed. In the context of synchrotron nanometrology, extreme ultraviolet (EUV) and x-ray scattering is used to characterize the structure of nanosystems. For instance, the pattern recorded in the grazing-incidence small-angle scattering experiments can be used to reconstruct the shape of lamellar gratings [5] with precision on the subnanometer scale. It was also experimentally shown that the diffuse scattering pattern encodes the statistical parameters of the grating's wall roughness (line edge/width roughness) [6,7]. Thus, scattering can be used not only to characterize the averaged parameters of the nanostructure but also to inspect the statistics of its surface imperfections. EUV scatterometry has its main applications in microelectronics because the mirrors used in EUV lithography today are based on Mo/Si multilayer systems optimized for maximum reflectivity at 13.5 nm. Such research is currently of great interest, and we see three distinct reasons for this. First, the commercial success of the EUV lithography. Second, recent developments in high-harmonic EUV generation [8,9] may allow the EUV scatterometry in-lab, which until now has mainly been possible only on synchrotrons. Finally, the third reason is that modern elements of microelectronics, like gate-all-around transistors, have a complex architecture that necessitates an advanced three-dimensional (3D) metrology. Such sophisticated manufacturing technology requires modeling of the scattering on the complex shape structures, which in turn motivates improvement of the computational schemes. This describes just one use case in which solving the wave scattering problem is essential, and there are many more.

We focus on solving the wave equation for a periodic scattering potential. Several theoretical and numerical approaches exist for solving this type of problem. Among them are the finite-element method [10], the finite-difference time domain approach [11], methods based on boundary integral equations [12], surface coordinate transform (C method) [13], volume integral equation method in curvilinear coordinates [14], and the Fourier modal methods. Fourier modal methods exploit the periodicity of a structure's scattering potential to formulate the solution in terms of the sum of the plane waves. As mentioned, the problem of scattering on a periodic potential arises in different fields of science. Thereby, there are several instances of Fourier modal theories, which historically developed in parallel with little to no overlap in the literature. The scientific community around diffractive optics and holography uses the rigorous coupled wave analysis (RCWA) theory [15]. The synchrotron radiation community employs the dynamic diffraction theory (DDT) [16,17]. Essentially, these theories are identical. The central idea is to represent the scattering potential as the discrete Fourier series and then to search for the solution in terms of the Bloch series. This allows the reduction of the wave equation, which is inconveniently a partial differential equation, to a system of ordinary differential equations, which one can solve in terms of the eigenvalue problem. This idea constitutes both the RCWA and the DDT. However, there is a difference in how these theories are developed, which is due to the specifics of each of the application fields. For example, constructing a Bloch series of only two components, a reflected wave and a diffraction order of interest is often sufficient in the x-ray range. Indeed, the wavelength for x-rays is shorter than in the optical range, and therefore, a phase difference between different plane waves is higher. Hence, interference between the diffraction orders is a rare occurrence for x-rays. In contrast, in the optical range one typically has to consider the multiple wave interference to solve the wave equation with satisfying accuracy. That is why RCWA is formulated to account for

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an arbitrary number of wave modes, whereas in DDT, one typically finds an explicit solution for each wave mode. On the other hand, the explicit form of DDT allows us to account for any orientation of periodic media, while RCWA assumes only a laterally periodic structure. Again, this difference is driven by applications: diffraction gratings that are periodic along the surface of the structure for RCWA, and crystals periodic in three dimensions for DDT. In RCWA, the complex line profile of gratings is approximated with a series of layers, homogeneous in the vertical direction. Then, the solution in each layer is chosen so that the particular solution across the whole structure is continuous.

In summary, RCWA and DDT are similar theories. The RCWA considers many-wave interference but works only on vertically homogeneous structures, while DDT considers structures with an arbitrarily oriented periodicity. The similarity between these approaches was mentioned in early works [15]. Therefore, the idea of combining the two approaches, so that both multiple wave interference and complex structure are within reach, is a natural one. One way to implement this was first demonstrated in Ref. [18] and further developed in Ref. [19]. In considerably recent research [20], the idea of unifying both theories was revisited. In addition, a new idea for treating the diffraction gratings with complex profiles without slicing them into vertically homogeneous layers was proposed [20].

With this article, we aim to once again draw the attention of the optical and the synchrotron scientific communities to the fundamental similarity of RCWA and DDT. Based on the ideas in Refs. [18–22], we construct yet another unified Fourier modal theory. We do this in a simplistic manner, which allows us to focus on the main ideas essential for accounting for multiple wave interference in structures with complex profiles. For this theory, we derive the generalized boundary conditions in linear algebraic terms so that it would be possible to implement them numerically in a computationally efficient way in the future. Finally, we implement a polygonal approach to approximate the shape of grating profiles. This opens a direction for future research toward model-independent simulations of diffraction gratings. We test our approach against the conventional RCWA and the finite-element method. We also test the approach to analyze actual experimental data previously published in Ref. [5]. The numerical and experimental models presented in this article are relevant for the further development of EUV or soft-x-ray nanometrology. However, it should be emphasized that the actual applications go beyond this framework.

II. THEORY

First, let us describe the mathematical notation used in this article. Fourier modal methods describe waves in terms of amplitude vectors, which are vectors of Hilbert space. Amplitude vectors and matrices are written in capital letters in normal font, for example, A. The space vectors, which describe the real geometric objects, are written in lowercase bold letters, for example, $\boldsymbol{r}$, to avoid confusion with amplitude vectors. The only exception to the capital letter notation is the electric field. The notation for the electric field is $E(\boldsymbol{r})$, which is well established in the literature. The Fourier components of $E(\boldsymbol{r})$ are written as $\hat{E}_g$ and they constitute the amplitude vector E. The electric field in real space, $E(\boldsymbol{r})$, is always explicitly written as a function of coordinate to avoid confusing it with the amplitude vector E. In this article, we neglect polarization effects, so $E(\boldsymbol{r})$ is a scalar field.

We solve the diffraction problem for nanoscale gratings. We neglect the crystallinity of the material and consider it as a continuous medium with a dielectric susceptibility χ. The only periodicity in this structure is therefore the period of the grating D. We choose the reference frame so that the lines of the grating are perpendicular to the x axis and parallel to the y axis. Thus, the surface of the sample lies in the xy plane and the z axis is perpendicular to the surface. The explicit representation of the wave vector of the incident beam $\boldsymbol{k}_i$ can be chosen arbitrarily. However, we find it more illustrative to define it as $\boldsymbol{k}_i = k[\cos\alpha_i \sin\phi;\ \cos\alpha_i \cos\phi;\ -\sin\alpha_i]^T$, where α_i is the grazing incidence angle and ϕ is the azimuthal angle, so that $\phi = 0^\circ$ corresponds to conical diffraction, i.e., the diffraction geometry where the incident beam lies in the plane perpendicular to the lines of the grating. We start with the Helmholtz equation in the scalar field approximation:

$$(\Delta + k^2)E(\boldsymbol{r}) = -k^2\chi(\boldsymbol{r})E(\boldsymbol{r}). \tag{1}$$

The scalar approximation is usually valid in the x-ray and EUV region for sufficiently small grazing incidence angles. The validity of the approximation depends on the structure of the scattering potential, see validity criteria in [23,24]. Here, k is the wave number in the vacuum and χ is the dielectric susceptibility, the term $-k^2\chi$, represents the scattering potential in the case of the photons. The dielectric susceptibility function is linearly dependent on the electron-density distribution, thus it encodes the geometry of the nanostructure. Since we consider periodic nanostructures, $\chi(\boldsymbol{r})$ can be represented as a Fourier series:

$$\chi(\boldsymbol{r}) = \sum_g \chi_g e^{i\boldsymbol{g}\cdot\boldsymbol{r}}. \tag{2}$$

For brevity, we allow ourselves an ambiguous use of the index notation: $\boldsymbol{g}$ is the reciprocal space vector. The subscript g in the sum in Eq. (2) implies the sum over all nodes in the reciprocal space. In general, we will consider a two-dimensional periodic structure $\boldsymbol{g} = [2\pi m/d_x,\ 2\pi n/d_z]^T$, where $m, n \in \mathbb{Z}$ and $d_{x,z}$ are the elementary cell sizes.

The solution of a wave equation with a periodic scattering potential can be expressed as a series of plane waves by the Bloch theorem. Thus, an ansatz for Eq. (1) is

$$E(\boldsymbol{r}) = \sum_g \hat{E}_g e^{i\boldsymbol{k}_g\cdot\boldsymbol{r}}. \tag{3}$$

In the far field (in the ambient at a sufficient distance from the surface of the sample), the phases of these plane waves are related to the nodes in reciprocal space $\{\boldsymbol{g}\}$ by the Laue diffraction condition $\boldsymbol{k}_g = \boldsymbol{k} + \boldsymbol{g}$, where $\boldsymbol{k}$ is the wave-vector of the incident rays in vacuum. Thus, $\boldsymbol{k}_g$ is the wave vector for the gth wave mode (or gth diffraction order). Direct substitution of Eqs. (3) and (2) in the Eq. (1) yields a system of linear equations:

$$\left(k_g^2 - k^2\right)\hat{E}_g = k^2 \sum_h \chi_{g-h}\hat{E}_h. \tag{4}$$

This is an infinite system of equations, for each node denoted by g each such equation contains infinite Fourier components of χ. The system is derived with use of the phase shift property of the Fourier series: $\chi_{g-h} = \chi_g e^{i\boldsymbol{hr}}$, for any arbitrary constant $\boldsymbol{r}$.

The physical meaning of Eq. (4) becomes apparent if we rearrange it. In the sum on the right-hand side of Eq. (4), consider the term $\chi_0 \hat{E}_g$ that corresponds to $g = h$. Let us move it to the left-hand side. Then, by introducing $\kappa^2 = (1 + \chi_0)k^2$, one can rewrite Eq. (4) as follows:

$$\left(k_g^2 - \kappa^2\right)\hat{E}_g = k^2 \sum_{h \neq g} \chi_{g-h}\hat{E}_h, \tag{5}$$

The Fourier component χ_0 represents the average dielectric susceptibility of the structured medium. Therefore, κ can be interpreted as the wave number within the structure as if it were an unstructured continuous medium (effective homogeneous medium). If the wavenumber k_g of a diffracted wave $k_g = \kappa$, then for higher orders $g \neq 0$ and Eq. (5) has only trivial solutions $\hat{E}_g = 0$. This has two consequences. First is a trivial one: if the medium is indeed homogeneous, then $k_g = \kappa$ and there is only a specularly reflected wave $\hat{E}_0$. Second, in the structured medium, diffraction is only possible if k_g deviates from κ. Thus, Eq. (4) describes dispersion in the structured medium.

To calculate the dispersion, the symmetry of the scattering potential can be taken into account: all wavevectors should be invariant with respect to translations along the surface: $\boldsymbol{k}_{g\parallel} = \boldsymbol{k}_{0\parallel} + \boldsymbol{g}_{\parallel} \equiv \text{const}$. This translational invariance is due to the symmetry in the scattering potential $-k^2\chi(r)$. Namely, the grating is bounded from the top by the vacuum and from the bottom by the semi-infinite substrate. Along these boundaries there is no gradient of the potential, so the projection of the wave vector onto these boundaries should be constant. A somewhat analogous line of reasoning is given in Ref. [25] on page 16, where the motion of a particle is considered in terms of classical Lagrangian mechanics. Then, the aberration of k_g is only due to a change in its vertical component k_{gz} [26] for which we can write $(k_g^2 - k^2) = (k_{gz} + g_z)^2 - \kappa_{gz}^2$, where

$$\kappa_{gz}^2 = (1 + \chi_0)k^2 - (\boldsymbol{k}_{\parallel} + \boldsymbol{g}_{\parallel})^2, \tag{6}$$

and the k_{gz}^2 is a value to be sought. Equation (6) is the equation of a sphere and it describes the dispersion in the effective medium (spherical dispersion equation). In the trivial case of homogeneous media, it results in the Snell's law. Now, introducing normalization $\xi = k_{gz}/k$, we write Eq. (4) as

$$\left(\xi^2 + \frac{2g_z}{k}\xi + \frac{g_z^2 - \kappa_{gz}^2}{k^2}\right)\hat{E}_g = \sum_h \chi_{g-h}\hat{E}_h. \tag{7}$$

This is an infinite system of quadratic equations with respect to ξ with infinite number of terms on the right-hand side. We note once again that ξ corresponds to the vertical components of the wave vectors. Note that the index g runs over an equation in a system and the index h corresponds to the term in the sum. We can give a physical interpretation to this: the index g corresponds to a wave mode $\hat{E}_g$, and h enumerates a vector from the basis (an eigenstate vector), a linear combination of which forms $\hat{E}_g$ itself. In these linear combinations, the components χ_{g-h} play the role of coupling coefficients. Since $\chi(\boldsymbol{r})$ is $\mathbb{L}^2$ integrable, the larger the separation $|g - h|$, the smaller the coupling. It is more convenient to analyze such systems using linear algebra.

In the language of linear algebra, relations in Eq. (7) take the form

$$[\xi^2 I + \xi A + B]E = 0. \tag{8}$$

Here, I is the identity matrix, $A = \text{diag}(\ldots 2g_z/k \ldots)$ and $B = \text{diag}(\ldots [g_z^2 - \kappa_{gz}^2]/k^2 \ldots) - C_{[\chi]}$. Last term here is a Toeplitz matrix [27] filled with the Fourier components χ_g:

$$C_{[\chi]} = \begin{bmatrix} \ddots & & & & \\ & \chi_0 & \chi_{\bar{1}} & \chi_{\bar{2}} & \\ & \chi_1 & \chi_0 & \chi_{\bar{1}} & \\ & \chi_2 & \chi_1 & \chi_0 & \\ & & & & \ddots \end{bmatrix}. \tag{9}$$

Note again that we are considering a two-dimensional (2D) structure, and each component χ_g corresponds to a node g in the reciprocal space which inherently has two coordinates, but for the brevity of notation we use the single index g. To construct the matrix $C_{[\chi]}$, a 2D grid of nodes $\{\boldsymbol{g}\}$ is unfolded in a one-dimensional (1D) array in any arbitrary order, it is only important to maintain same ordering for the other matrices involved in computations. A Toeplitz matrix $C_{[\chi]}$ encodes information about the scattering potential: the more Fourier components, i.e., more nodes in reciprocal space are taken into account, the more detailed the structure is obtained. Let n be the number of nodes, so that $C_{[\chi]} \in \mathbb{C}^{n\times n}$. The matrix A contains vertical coordinates g_z of the nodes in the reciprocal space, the matrix B encodes the scattering potential structure, and its diagonal part takes into account the spherical dispersion. Equation (8) represents dispersion inside the structured medium. It has the form of a quadratic eigenvalue problem (QEP) with respect to the eigenvalue ξ and eigenvector E.

Consider the case of a vertically homogeneous structure: for instance, a box-shaped grating. The structure is constant with respect to the vertical translation, so there are no reciprocal space nodes along g_z and, hence, $A \equiv 0$. Thus, the QEP in Eq. (8) reduces to the ordinary eigenvalue problem with respect to ξ^2. Then each eigenstate has two symmetric solutions with $\pm\xi$, and the near-field is then represented by the series of standing waves, i.e., each wave is a sum of two plane waves with an opposite phase. This reduces the problem to the classical RCWA theory. To analyze an actual grating, which is not necessarily homogeneous along the vertical direction, with the RCWA, it must be approximated by box-shaped gratings stacked in layers. The dispersion equation can then be solved separately in each layer. Finally, the solutions in each layer can be stitched together into $E(\boldsymbol{r})$ by solving the boundary-value problem (BVP) at each interface between layers.

An alternative idea is proposed in Ref. [20]. Instead of dividing the structure into layers, one can consider the structure that is periodic in one direction as a 2D periodic structure, where periodicity in the vertical direction takes up just a single period (see Fig. 1). This allows us to take into account the structure change in the vertical direction by using the 2D Fourier transform [cf. Eq. (2)], so that the BVP has to be

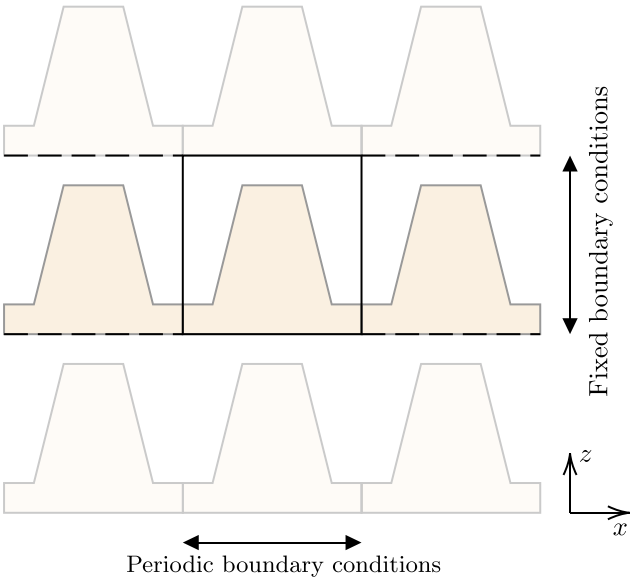


FIG. 1. A sketch illustrating the idea from Ref. [20]: The lamellar 1D lattice can be represented as a 2D photonic crystal that occupies exactly one period in the vertical direction, while maintaining periodicity along the lateral direction.

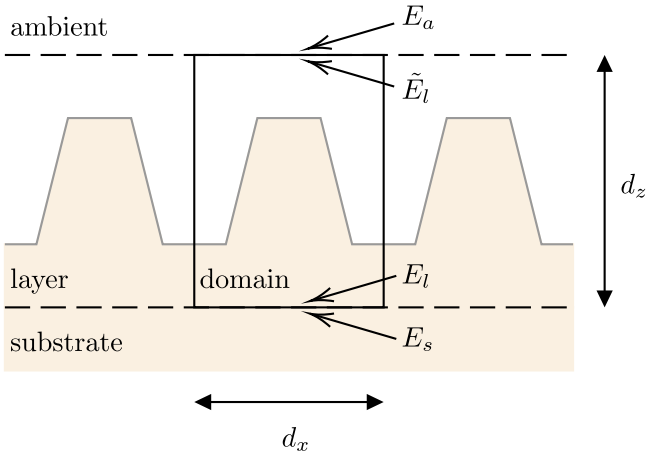


FIG. 2. A sketch illustrating the structure of the BVP. The system is divided into three parts: ambient, structured layer, and substrate. The arrows indicate which part the amplitude vectors E refer to. The near field is sought within a computational domain.

solved only for the substrate and the ambient. However, this violates the $\pm\xi$ symmetry, and one now has to deal with QEP.

Nevertheless, the QEP can be solved by an elegant linearization. In the context of x-ray diffraction simulations, this was shown in Ref. [18]. Let $D = \xi E$, then Eq. (8) becomes $\xi D + AD + BE = 0$, and these two equations can now be written in a block matrix form

$$\begin{bmatrix} 0 & I \\ -B & -A \end{bmatrix} \begin{bmatrix} E \\ D \end{bmatrix} = \xi \begin{bmatrix} E \\ D \end{bmatrix}. \tag{10}$$

Thus the problem is linearized into an eigenvalue problem, with eigenvectors $[E\ D]^T$ and eigenvalues ξ. The top half of an eigenvector represent an eigenstate, and amplitudes $\hat{E}_g$ in Eq. (3) can be calculated as a linear combination of the eigenstates.

The next step is to solve the BVP. Unlike the RCWA, the solution is quite simple, since there are only two interfaces to consider. There is an interface between the ambient and the structured layer, and between the layer and the substrate. To do so, we consider four amplitude vectors: E_a is the representation of the field in the ambient directly at the surface of the structure, $\tilde{E}_l$ is also for the field at the surface but inside the structured layer. Correspondingly, E_l is the field at the bottom of the structure and E_s is in the substrate (see Fig. 2). The BVP is solved by imposing the continuity on the amplitudes and their gradients along the vertical direction at the interfaces. This can be expressed as

$$M_a E_a = M_l \tilde{E}_l, \quad M_l E_l = M_s E_s, \tag{11}$$

where M_a and M_s describe the refraction in the homogeneous media, that is, the ambient and the substrate, respectively, and M_l corresponds to the structured layer.

The matrix M has the simplest form if the following order of the components of the amplitude vector in homogeneous media is chosen: the amplitude vectors are divided into two blocks $E = [T\ R]^T$, where R consists of the amplitudes of upward propagating plane waves (decay modes with respect to the direction of z) and T is for downward propagating waves (gain modes). Moreover, the order within R and T is such that nth elements of each form a standing wave with an opposite phase: $T_n e^{-ik_{zn}} + R_n e^{+ik_{zn}}$. Then, for M we can write

$$M_{a,s} = \begin{bmatrix} I & I \\ -K & K \end{bmatrix}, \tag{12}$$

where $K = \mathrm{diag}(\ldots \kappa_{0z} \ldots)$. The vertical components κ_{0z} are calculated using the spherical dispersion Eq. (6), choosing appropriate dielectric constants χ_0 for each matrix (ambient and substrate). The upper half of a matrix corresponds to the continuity of the field itself, and the lower half to its gradient, which for the plane waves simply scales the field with the wave number, hence Eq. (12).

As described above in the structured medium, the amplitude vectors are the linear combinations of the eigenstates, i.e., the eigenvectors E, computed by solving QEP. Let us bundle the eigenvalues ξ into a vector Ξ and the eigenvectors into a matrix F. Note that we consider the eigenvectors as columns. Then, for the matrix M_l we derive:

$$M_l = \begin{bmatrix} F \\ (G \oplus k\Xi) \circ F \end{bmatrix}. \tag{13}$$

Here, symbol $\oplus$ denotes the outer sum $(a \oplus b)_{ij} = a_i + b_j$, and G is the vector composed of vertical coordinates g_z of nodes in the reciprocal space. Again, the upper half of the matrix M_l corresponds to the field continuity and the lower half to the gradient continuity. The matrix $G \oplus k\Xi$ is composed of the vertical components of the wave vectors and, unlike the homogeneous case, is not diagonal, which takes multiple wave interference into account. The symbol $\circ$ denotes the Hadamar (element-wise) matrix multiplication.

The next step in solving the BVP is to link the amplitudes at the surface of the structured layer and at the bottom. This is done using a unitary matrix that takes into account the phase

change and absorption of the propagating waves:

$$\tilde{E}_l = QE_l, \tag{14}$$

where $Q = \mathrm{diag}(e^{-ik\Xi d_z})$. To derive the far-field solution, one can solve Eqs. (11) and (14) with respect to R_a (the second block of E_a). The resulting elements of R_a represent the diffraction amplitudes with the corresponding order. The near-field solution can be obtained by solving the equations with respect to E_l and subsequently applying the Fourier transform to obtain the spatial profile of the electric field, $E(\boldsymbol{r})$. For that, one can assume that $T_a = (\ldots 0, 1, 0 \ldots)$, which corresponds to the incident plane wave: amplitudes of all higher-order wave modes are zero. Further, one assumes $R_s = (\ldots 0, 0, 0 \ldots)$ because the reflection from the bottom of the substrate is negligible. Thus, from a mathematical standpoint, the problem is solved.

Unfortunately, it is practically impossible to find a numerical solution in this manner. This is because the matrix Q is ill-conditioned because it contains both exponentially increasing and decaying elements. By backpropagating amplitudes from values below the machine epsilon, one simply calculates computation noise instead of the amplitude values. To mitigate this problem, one has to find a form of Eqs. (11) and (14) that includes only decaying exponents. This idea is known and widely applied. For instance, in the context of x-ray diffraction, it is discussed in Ref. [21].

Let us now implement this idea for our problem. Generally, the order at which the eigenvalues ξ are sorted is arbitrary, but one needs to separate gain and decay wave modes. Therefore, we sort and split the vector ξ of eigenvalues into a block vector $\Xi = [\Xi_- \ \Xi_+]^T$, such that Ξ_- contains eigenvalues with $\mathrm{Im}\,\xi < 0$ (gain modes) and Ξ_+ contains $\mathrm{Im}\,\xi > 0$ (decay modes). Let us assume that the number of decay modes is not the same as the number of gain modes. Such an assumption is incorrect because the dispersion is described by a quadratic equation. However, it can occur numerically. The sequence of column eigenvectors in F must be sorted accordingly. Furthermore, one should rewrite Q as a block matrix as well:

$$Q = \begin{bmatrix} Q_- & 0 \\ 0 & Q_+ \end{bmatrix}. \tag{15}$$

Now we can rewrite Eqs. (11) and (14) in the block matrix form:

$$\begin{bmatrix} T_l \\ R_a \end{bmatrix} = P_{al} \begin{bmatrix} T_a \\ R_l \end{bmatrix}, \quad \begin{bmatrix} T_s \\ R_l \end{bmatrix} = P_{ls} \begin{bmatrix} T_l \\ R_s \end{bmatrix}. \tag{16}$$

Notice that here the order of propagation is changed: each amplitude vector contains amplitudes from different media. Now, using block-matrix algebra, one can easily find an explicit expression for both P_{al} and P_{ls}:

$$P = \begin{bmatrix} Q_-^* V_{11}^\dagger & -Q_-^* V_{11}^\dagger V_{12} Q_+ \\ V_{21} V_{11}^\dagger & V_{22} Q_+ - V_{21} V_{11}^\dagger V_{12} Q_+ \end{bmatrix}. \tag{17}$$

Here, the "†" superscript denotes the Moore-Penrose inverse (or pseudoinverse), which is a generalization of inversion for nonsquare matrices; V is the 2×2 block matrix with blocks V_{ij}. For P_{al}, one has to take $V^{(al)} = M_a^{-1} M_l$. For P_{ls}, one has to take $V^{(ls)} = M_l^\dagger M_s$ and to set $Q = I$, this is because we do not consider the propagation inside the substrate. Again, this is because we no longer rely on there being the same number of decay modes as there are gain modes. Therefore, the sizes of the blocks in Eq. (16) are different. See the Fig. 3 for an illustration. Note that the matrix P does not explicitly contain Q_-. It contains only its inversion, which, being a unitary matrix, can be computed as $Q_-^{-1} = Q_-^*$. Now the Q_-^* matrix also contains decaying exponents. This eliminates the problem of backpropagating values below the machine epsilon.

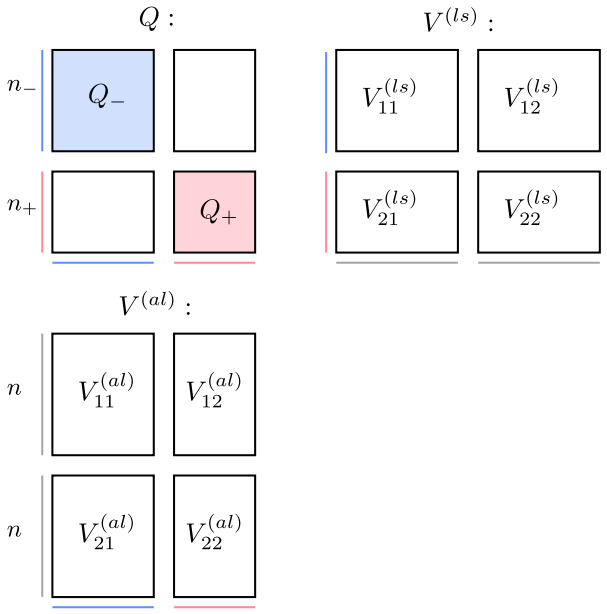


FIG. 3. Block matrix diagram for Eq. (16). The size of the matrices is shown schematically. Here N is the number of wave modes considered in the ansatz, n_+ is the number of gain modes, n_- is the number of decay modes.

Now the amplitude coefficients can be determined as

$$T_l = \left(I - A_{12}^{(al)} A_{21}^{(gs)}\right)^{-1} A_{11}^{(al)} T_a, \tag{18}$$

and

$$R_l = \left(I - A_{21}^{(ls)} A_{12}^{(al)}\right)^{-1} A_{21}^{(ls)} A_{11}^{(al)} T_a. \tag{19}$$

Having computed the vectors T_l and R_l, one can find $E(\boldsymbol{r})$ simply by applying the fast Fourier transform to solve Eq. (3). Thus, the near-field problem is solved. In this framework, one solves the far-field problem by finding the transmission T_s and reflection R_a amplitudes:

$$T_s = A_{11}^{(ls)} T_l, \quad R_a = A_{21}^{(al)} T_a + A_{22}^{(al)} R_l. \tag{20}$$

By obtaining these amplitude vectors, the problem is solved.

The last theoretical consideration of this work is the model of the structure itself. To solve the near-field problem, one needs to represent the structure in terms of the Fourier transform in Eq. (2). We propose to approximate the grating profile with a polygon. The Fourier transform of the indicator function γ of a polygon [$\gamma(\boldsymbol{r}) = 1$ if $\boldsymbol{r} \in \Gamma$ and $\gamma(\boldsymbol{r}) = 0$ otherwise] has a simple closed-form solution [28]. In Ref. [29], the numerically stable solution was proposed to eliminate divergence near $Q = 0$. However, we do not want to deal with such problems since we consider our structure to be periodic, so we only compute the transform of χ on a discrete grid,

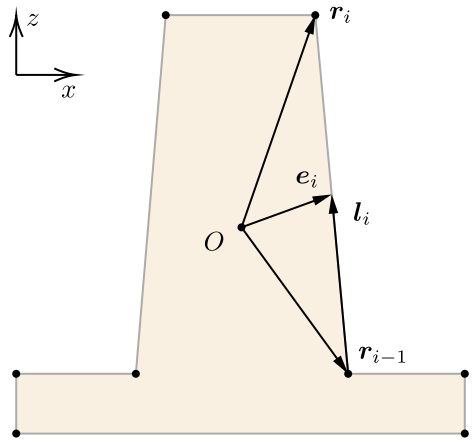


FIG. 4. A sketch of a nonself-intersecting polygon Γ. The origin O can be chosen arbitrarily. The vertices $\boldsymbol{r}_i$ are indexed counterclockwise. The vectors $\boldsymbol{e}_i$ and $\boldsymbol{l}_i$ are involved in Eq. (22).

so our only problem is to get rid of the singularity at exactly $Q = 0$.

Consider a nonself-intersecting polygon Γ with the indicator γ. By definition, the Fourier transform of γ is

$$\hat{\gamma}(\boldsymbol{q}) = \int_{\Gamma} d^2 r e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}. \tag{21}$$

Let a polygon be described with a set of vertices with coordinates $\{\boldsymbol{r}_i\}$ (see Fig. 4). The set $\{\boldsymbol{r}_i\}$ the set is ordered, such that the vertices are in the counterclockwise sequence. By virtue of the Stokes theorem, the integration in Eq. (21) can be turned into the integration over the line segments of the polygon Γ. See the detailed proof in Ref. [29]. The integration over the line segments yields

$$\hat{\gamma}(\boldsymbol{q}) = \sum_j \frac{q_y l_{yj} - q_x l_{xj}}{q} \{\mathrm{sinc}(-\boldsymbol{q}\cdot\boldsymbol{l}_j) e^{-i\boldsymbol{q}\cdot\boldsymbol{e}_j} - 1\}, \tag{22}$$

with $\boldsymbol{e}_j = (\boldsymbol{r}_j + \boldsymbol{r}_{j-1})/2$ and $\boldsymbol{l}_j = (\boldsymbol{r}_j - \boldsymbol{r}_{j-1})/2$.

The described theoretical considerations constitute the polygonal approach. Here is the numeric recipe for calculating the diffraction on the grating. Start by calculating the dielectric susceptibility constant χ for the material for a given incident photon energy using tabular data. Create a polygon model of the grating's line profile by specifying the coordinates of each vertex and putting them into the sequence in counterclockwise order. Calculate the Fourier components χ_g by calculating the $\chi\hat{\gamma}(\boldsymbol{g})$ at all considered reciprocal space nodes $\{\boldsymbol{g}\}$, where χ is the dielectric constant of the material. Furthermore, put these components χ_g in the Toeplitz matrix Eq. (9). Note that the components χ_g are calculated on a two-dimensional grid, but to compose a Toeplitz matrix, one has to reorder them into a one-dimensional array. The order of that array is arbitrary, but it will define the order in the amplitude vectors R and T. Solve the linearized QEP [Eq. (8)] using any appropriate eigenvalue solver. Sort the eigenvalues into two sets: $\mathrm{Im}\,\xi > 0$ for decay modes and $\mathrm{Im}\,\xi < 0$ for gain modes. Sort column vectors in the eigenmatrix accordingly. The upper half block of the eigenvector matrix is the matrix F. Using this matrix, compute the propagation matrix M_l in Eq. (13). Using Eqs. (16)–(20) compute amplitude vectors. Amplitude R_a describes the far field of the scattered wave. To compute the near field, solve Eq. (3) through a fast Fourier transform (FFT) algorithm.

Thus, the problem of diffraction on periodically patterned surfaces can be solved. Once again, we would like to emphasize that this approach is based on the following ideas: First, the characteristic equation is reformulated as a quadratic eigenvalue problem [18], which can be solved using linearization [19]. Second, the structure on the patterned surface is considered to be a vertically periodic structure with a thickness that exactly matches one period of the structure [20]. Third, the structure factor is calculated using a simple equation for a polygon [29]. This provides a convenient formalism for modeling diffraction in gratings with arbitrarily shaped line profiles. This approach is based on the scalar approximation. The full vectorial characteristic equations, which account for the polarization state of the incident photons, are described in Ref. [20]. Through this work, we are extending this approach with equations that implement boundary conditions in a numerically stable manner. In practice, these are absolutely necessary for the simulations because increasing the number of Fourier modes in the z direction results in many eigenstates with amplitudes close to zero. Straightforward propagation of such amplitudes limits the precision of the computation by amplifying the machine noise. The boundary-condition equations are formulated so that the classical mount (coplanar diffraction), the conical mount, and all intermediate geometries (tilted around the surface normal at an arbitrary angle ϕ) can be considered. The only limitation here is that this is done in the scalar approximation. In this regard, it would be interesting to test how this formalism compares to the full vectorial simulations. Does it account for asymmetric, nonconical illumination geometry? How well does the formalism work for angles of incidence greater than the critical angle? We address these questions in the following section.

III. NUMERIC SIMULATIONS AND DATA ANALYSIS

The numerical examples in this section are performed in the context of the grazing-incidence small-angle x-ray scattering (GISAXS) experiments. In GISAXS, the sample is irradiated with a monochromatic x-ray beam under the grazing incidence conditions in which the angle of incidence is chosen near the critical angle (1 to 5 critical angles). Thus, the scattering geometry is fully determined by the two angles: the angle of incidence, α, and the angle ϕ defined as the angle between the grating lines and the lateral projection of the incident beam. Under these grazing incidence conditions, evanescent wave diffraction occurs. This physical phenomenon can be represented mathematically in two ways. First, a standing wave with a complex shape forms in the near field, $E(r)$. Second, bright-intensity diffraction peaks form in the far field. These peaks can be measured in a GISAXS experiment [30] and their integrated intensities are proportional to the squared absolute values of the amplitudes, $|R|^2$. This is analogous to r and k representations in quantum mechanics. We consider nanoscale lamellar gating to be a test structure. Then, we compare the near- and far-field calculations to those performed using other computational methods. Additionally, we compare the far-field calculations with the live

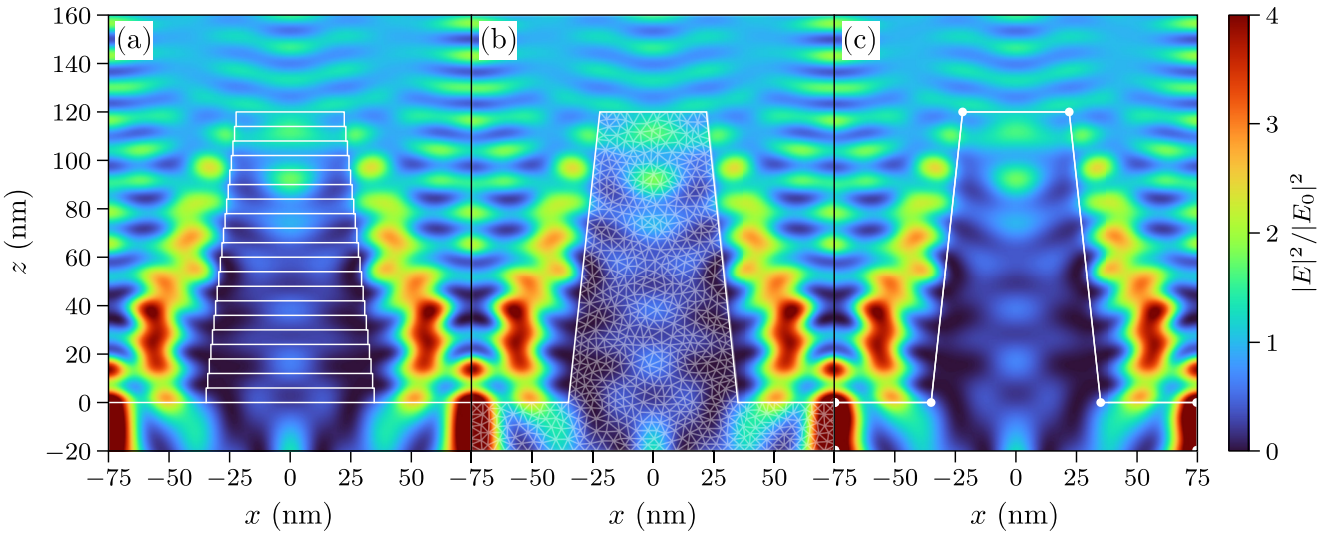


FIG. 5. Near field calculation comparison. Electric energy density $|E|^2$ (EED) calculated by (a) RCWA, (b) FEM and (c) polygonal approach. Normalized EEDs are plotted. The normalization is done such that the amplitude of the incident beam is strictly equal to unity. Test model: silicon grating with 150 nm lateral period, incident-beam photon energy $\hbar\omega = 5500$ eV, angle of incidence $\alpha_i = 0.5°$, conical mount ($\phi = 0°$).

experimental data measured at the bending magnet synchrotron radiation source.

We performed the near-field calculation using the conventional RCWA, the FEM, and the polygonal approach. The calculation is performed for the strictly conical geometry: $\phi = 0°$. The test structure model parameters are as follows: the lateral period (the pitch) $D = 150$ nm, the line height of the grating is $D = 120$ nm, and the sidewalls are inclined with an angle τ, $\tan\tau \approx 0.1$. The structure is irradiated by a monochromatic beam with the photon energy $\hbar\omega = 5500$ eV under the grazing incidence angle $\alpha = 0.5°$. The dielectric susceptibility χ is calculated for the pure silicon with nominal density $\rho = 2.33$ g/cm^3 according to the tabular data from Ref. [31]. The parameters are chosen to be comparable with an actual sample, which we consider further in the text [see its SEM image in Fig. 10(a)].

The results of all three simulations are shown in Fig. 5. The near-field solutions are represented as the normalized electric energy density (EED) $|E(r)|^2$. The EED is normalized such that the amplitude of the incident beam equals unity. As such, the EED is expressed as a unitless value. For the RCWA computation, the grating line is divided into 20 layers, and the number of Fourier modes taken for the ansatz is 41. These numbers of layers and modes appear to be sufficient, according to a convergence study. The ambient and the substrate are considered semi-infinite laterally homogeneous layers. The structure subdivision into layers is schematically shown over the electric energy density (EED) map in Fig. 5(a).

The FEM computation is done using the JCMWave software [10]. This package is often used as a reference for nanometrology because it is widely used in the relevant literature (see, among others, Refs. [32–34]) and has also been optimized for the very short x-ray wavelengths. Hence, we can consider the FEM computation as a benchmark. In the FEM, the computational domain is meshed into triangular elements or patches. In each patch, an approximation to the solution is sought as a linear combination of polynomial basis functions. This allows us to handle structures with complex shapes of line profiles. The finite element side boundary of $h = 6$ nm and a polynomial degree of $p = 4$ were used in the computation. The numerical precision in the simulation was verified until the quasi-exact result was achieved. The FEM results are shown in Fig. 5(b), together with the schematic representation of the computation mesh. The elements representing the ambient part of the computational domain are omitted for visual clarity.

Finally, we perform the computation with the polygonal approach using the numeric description given above. The computational domain of the same size as for the FEM computation is chosen, where a polygon with eight vertices represents the line and part of the substrate. In terms of nodes in reciprocal space, the computational domain size was chosen according to a convergence study. Computational accuracy increases with the number of nodes in the reciprocal space at the expense of computation time. Therefore, one aims to minimize the number of nodes while maintaining sufficient accuracy. To do this, we use the convergence study: the number of nodes is increased until the L_2 norm of the solution is saturated with a given accuracy. The size of 41×21 nodes appears to be sufficient for this calculation. The results are shown in Fig. 5(c).

By visual inspection, the results are in excellent qualitative agreement. Figure 6 shows the horizontal line cut of these solutions taken at the bottom ($z = 0$ nm), middle ($z = 60$ nm), and surface ($z = 120$ nm) of the structure, allowing one to inspect the difference in the simulations in more detail. Visually, the three solutions are indistinguishable. However, there is a difference in both absolute value and gradient, that can only be seen by zooming in (see the inset in Fig. 6). Specifically, the FEM and the polygonal approach results have a similar gradient in the chosen area, but different absolute values, and they both differ from the RCWA gradient. To compare the quality of results across the entire computational domain, we use relative L_2 norms in which the FEM results are taken as

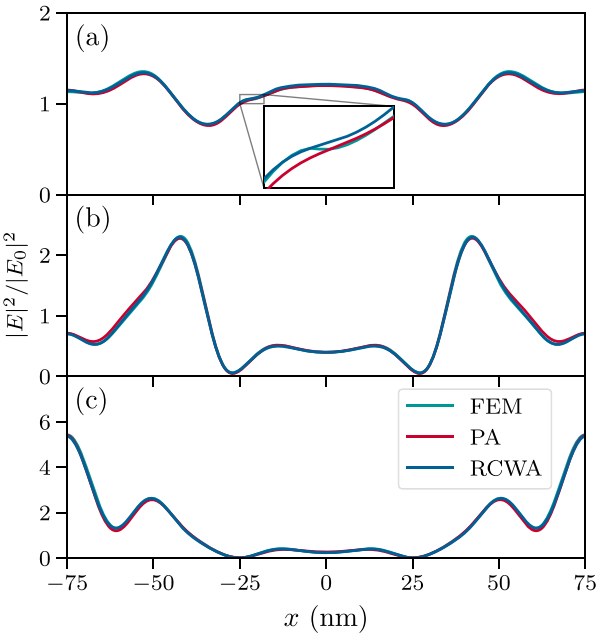


FIG. 6. The horizontal line cuts from the three (FEM, RCWA, and polygonal approach) near-field solutions (cf. Fig. 5) taken at the bottom ($z = 0$ nm), middle ($z = 60$ nm), and surface ($z = 120$ nm) of the structure.

the reference:

$$\frac{||\,|E| - |E_{\mathrm{FEM}}|\,||_2}{||\,|E_{\mathrm{FEM}}|\,||_2} = \frac{\sqrt{\sum |\,|E| - |E_{\mathrm{FEM}}|\,|^2}}{\sqrt{\sum |E_{\mathrm{FEM}}|^2}}.$$

Absolute values here compensate for an arbitrary choice of the incident-beam phase. The sum is taken over the points of arrays E. Such a norm for the difference between RCWA and FEM is 0.01, and between the polygonal approach and FEM is 0.02. Therefore, the polygonal approach provides an approximate near-field solution of the same order of accuracy as RCWA and FEM.

The comparison of the far-field solutions between FEM and polygonal approach is shown in Fig. 7. The previous near-field solutions were calculated for one fixed geometry fully defined by the angle of incidence α and the azimuth angle ϕ. In other words, a single solver run was required for each near-field solution. Figure 7 plots the far-field amplitudes with respect to different angles of incidence. Therefore, a separate solver run is required for each angle, making such simulations very computationally demanding. The simulations were performed for two grating orientations relative to the scattering plane. One orientation was strictly conical at $\phi = 0°$ [see Fig. 7(a)], and the other was tilted from conical at $\phi = 5°$ [see Fig. 7(b)]. The FEM simulation is a full vectorial simulation. As assumed, in this experimental setting, the difference in polarization state is negligible. The FEM provided equivalent solutions for s and p polarization at the level of precision of the numerical scheme. We chose to plot the p-polarization solutions because the scalar approximation is typically closer to the s-polarization solution. This emphasizes the discrepancy in the solutions for the worst-case scenario in such an experimental setting. For brevity, only three orders of diffraction (-1, 0, and $+1$) are plotted. The FEM solutions are plotted as

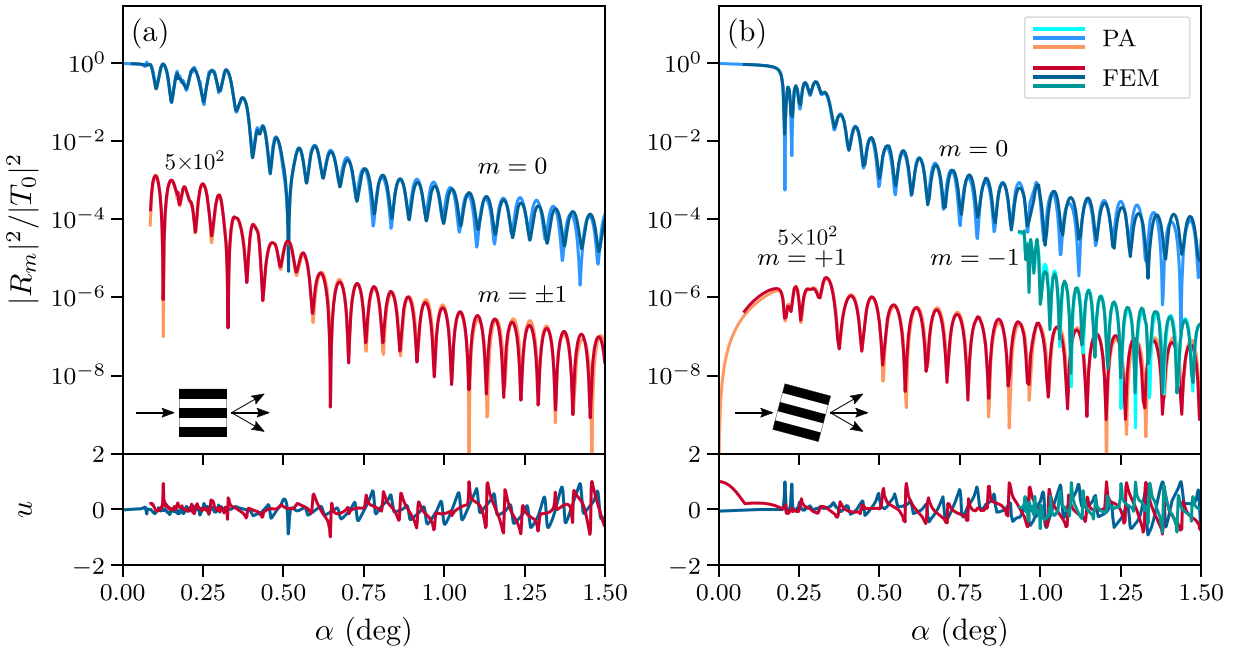


FIG. 7. Far-field calculation comparison. Calculations performed using FEM (red, green, and blue) solid lines and a polygonal approach (lines are marked with corresponding shades of color). FEM calculations are performed for p-polarization, whereas polygonal calculations are inherently scalar. Far-field solutions are represented by diffraction amplitudes $|R_m|^2$. These amplitudes are plotted against the angle of incidence α. The three orders of diffraction are plotted: $m = -1, 0, +1$. For visual clarity, the $m = \pm1$ curves are scaled down by a factor of 5×10^{-2}. (a) strictly conical geometry ($\phi = 0°$) (b) grating tilted around the surface normal ($\phi = 5°$). The lower section shows the relative residuals u between the FEM and polygonal approach. The residual curves are color-coded to relate to the corresponding order of diffraction.

solid red, green, and blue lines. The corresponding shades of these colors are used for the polygonal approach.

Due to symmetry, the $+1$ and -1 orders have equivalent amplitudes in strictly conical diffraction. In tilted geometry, however, they behave differently with respect to α. There is a simple physical interpretation for these curves. The critical angle for silicon at this photon energy is $\alpha_c \approx 0.33^\circ$. At angles smaller than the critical angle, one sees an irregular pattern that corresponds to the diffraction of evanescent waves. In other words, the grating also works as a waveguide. Therefore, there are oscillations; however, the intensity of diffraction does not decay as $\mathcal{I} \sim q_z^{-4}$. After the critical angle, the oscillations begin to decay and take on a regular structure. As the grating tilts away from a strictly conical orientation [see Fig. 7(b)], the range of angles at which the wave-guiding effect occurs becomes more narrow. Also, the symmetry between $+1$ and -1 orders breaks, and they appear on a diffraction cone at different angles α.

As can be seen in Fig. 7, the curves visually demonstrate good agreement between the FEM and the polygonal approach. Nevertheless, the discrepancies between the two methods are apparent. To emphasize the differences, we calculated the relative discrepancy,

$$u = \frac{|R_{\mathrm{FEM}}|^2 - |R_{\mathrm{PA}}|^2}{\max(|R_{\mathrm{FEM}}|^2, |R_{\mathrm{PA}}|^2)}.$$

These relative discrepancy curves u are plotted in the lower sections of Fig. 7. As one can see, the discrepancy curve becomes larger in value and begins to exhibit regularities at higher angles. These regularities indicate the phase discrepancies in the oscillations. This is likely caused by the truncation of vertical spatial frequencies under fixed boundary conditions in the polygonal approach.

Of course, the computational time depends on the settings for each algorithm: number of elements in the FEM, number of wave modes, number of reciprocal space nodes for RCWA and for the proposed approach. Therefore, comparing the computational efficiency of these algorithms is not a straightforward task. In our research, the RCWA computations were performed on the timescale of seconds, the polygon computation was performed on the timescale of tens of seconds and the FEM computation was performed within a few minutes. It is also important to note that in the state-of-the-art formulation of RCWA, the slanted edges of a grating can be taken into account, so we could achieve a faster solution with RCWA. If the structure becomes more complex, for example, if the profile of a grating's line becomes irregularly shaped, this advantage of RCWA would disappear. However, we could again use a modern formulation of the C method to account for such a structure with sharp edges [35]. Thus, our comparison of approaches should not be seen as a definitive comparison of computational speed. Nevertheless, our simulations show that a polygonal approach can show reasonable performance compared with the Fourier modal approach. In addition, for some systems, such as gratings with one material embedded in another, a polygonal approach may be more convenient.

Furthermore, we compare the polygonal approach simulations with the conical diffraction data measured in the GISAXS geometry, taken from a previously published research [5]. In Ref. [5], the structure parameters of an actual nanoscale grating were reconstructed by fitting the simulation to the data. A fixed set of geometric parameters, such as height, sidewall angle, pitch, etc., is used to describe the structure profile. Then, such a model was used for the forward diffraction simulation by the FEM. By repeating this process iteratively employing optimization algorithms, the best-fit set of parameters was found (for details, see the original study [5]). In this study, we focus on testing the polygonal approach, so the optimization procedure was not used. Instead, we take the best-fit solution from Ref. [5], construct a polygon from it, and perform the forward computation using the polygonal approach. In our simulations, we do not consider the divergence of the beam. This can be done by convolving multiple solutions with the instrumental function of the incident beam. It has been shown that the far-field solution is sensitive to the horizontal divergence on the scale of 0.1° [36]. Therefore, it is important to consider the divergence of the incident beam for quantitative reconstruction of a structure. However, in our study, we aim to validate the applicability of the polygonal approach for reconstruction where quantitative agreement is sufficient. This allows us to verify the applicability of the polygonal approach to analyzing actual nanostructures.

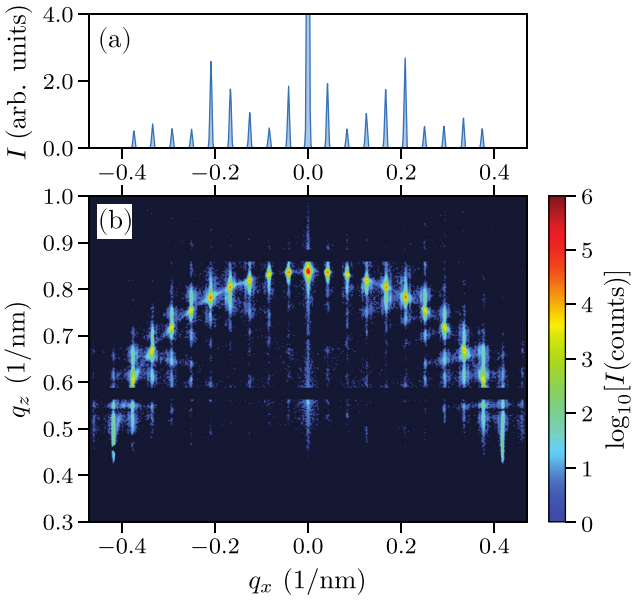


FIG. 8. (a) The GISAXS data of the silicon lamellar lattice originally published in Ref. [5]. Incident-beam photon energy $\hbar\omega = 5500$ eV, angle of incidence $\alpha_i = 0.86^\circ$, conical mount. (b) The scattering pattern is shown on a logarithmic scale. The line cut along the diffraction cone (a) is with respect to the reciprocal coordinate q_x. Integrals of the peaks in panel (a) are proportional to the corresponding diffraction efficiencies.

The GISAXS patterns are measured at the BESSY-II synchrotron at the four-crystal monochromator beamline [37]. The silicon grating sample with 150 nm pitch is prepared using electron-beam lithography. Scattering patterns were measured with a fixed grazing incidence angle of $\alpha_i = 0.86^\circ$ and at incident photon energy $\hbar\omega$ varying from 5500 to 5750 eV. A single GISAXS pattern for $\hbar\omega = 5500$ eV is shown in Fig. 8(b). Notice the bright diffraction peaks

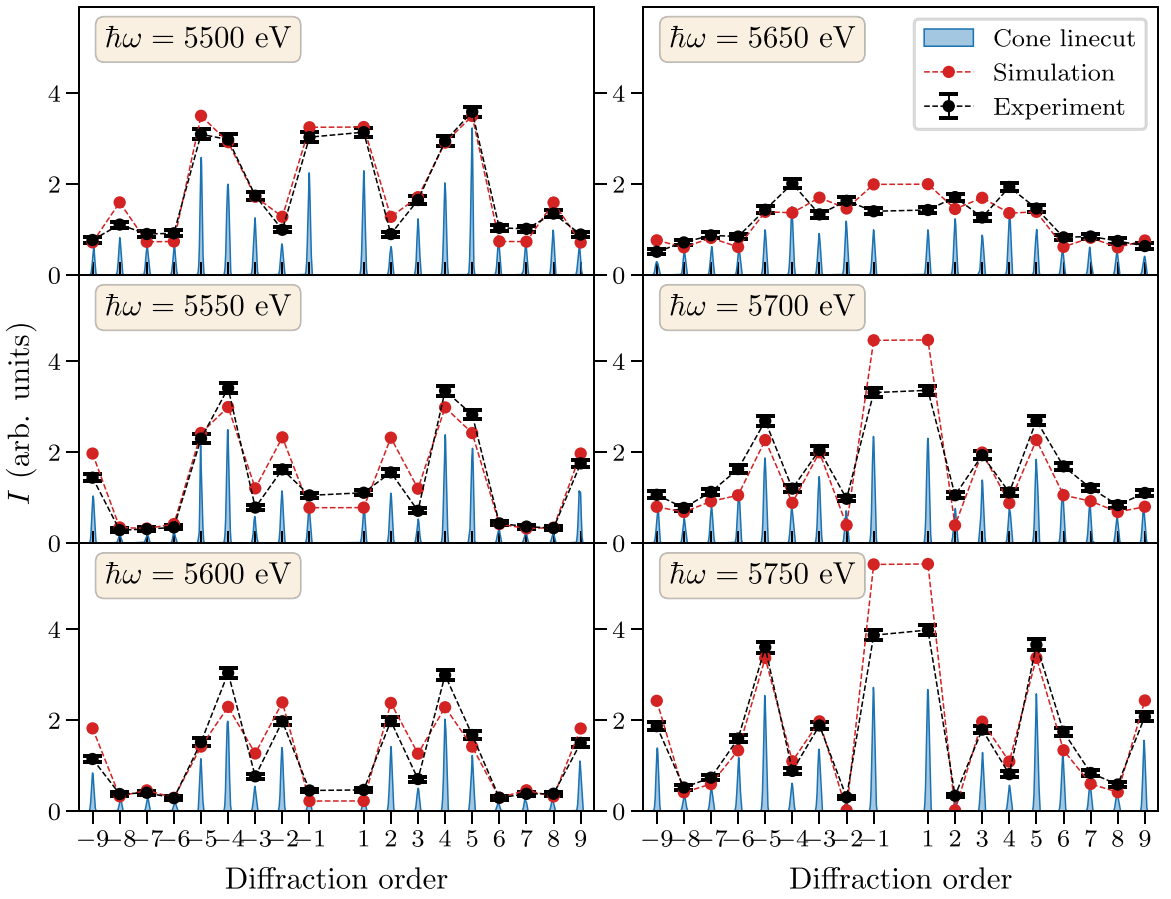


FIG. 9. Comparison of diffraction intensity extracted from GISAXS patterns (black markers) with the polygonal approach simulation (red markers). The corresponding line cuts along the diffraction cone are shown as blue lines (not to scale). Six measurements with photon energies ranging from 5500 to 5750 eV are considered.

situated around the circular section. The average structure of the grating's line defines the intensity distribution along these diffraction peaks. In order to extract the diffraction intensity distribution, we integrate the pattern along the q_y direction in a thin (seven pixels in width) stripe covering the cone section. The resulting line cut with respect to q_x is shown in Fig. 8(a). The final step in preparing the data is to integrate the cone section near the diffraction peaks and, thus, to obtain a set of the total diffraction intensities.

Carrying out this procedure, we extract the diffraction intensities from six GISAXS patterns measured for different $\hbar\omega$. Then, we simulate the diffraction with the polygonal approach. To do so, we construct a polygon by the best-fit model from. Note that keeping the counterclockwise order of vertices is crucial for the numerical algorithm. A polygon with 75 vertices is used for the simulations. The computational domain in the reciprocal space is 81×41 nodes. Amplitudes R_a [from the block-vector E_a in Eq. (11)] of the Bloch wave in the ambient above the sample surface were calculated using the polygonal approach.

$$\mathcal{I}_n = \frac{k_{zn}}{k_{z0}} |R_n|^2 \exp\left(-\frac{1}{2} q_x^2 \sigma^2\right).$$

To compare the results with the experiment, we take into account the flux normalization. The surface roughness was also taken into account, with the Debye-Waller damping factor. The value $\sigma = 1.87$ nm of the best-fit roughness amplitude was again taken from Ref. [5] and fixed.

The comparison of the forward simulation with the experimental data is shown in Fig. 9. The experimental data are shown with black markers connected with the dashed line for the visual aid. The error bars are calculated as a 3σ of the shot noise. Any other kinds of errors are not accounted for. The results of the simulation are in considerable agreement with the experimental data. The line profile (the upper part of a polygon used for the simulation) is shown against the SEM image [5] of the sample in Fig. 10(a). They are in good visual agreement. The near field corresponding to a $\hbar\omega = 5500$ eV and $\alpha_i = 0.86^\circ$ incidence is in Fig. 10(b).

Taken together, these results demonstrate the utility of the polygonal approach in simulating the diffraction of the synchrotron beam on nanoscale gratings in grazing-incidence geometry. As expected, the polarization effects are not prominent at grazing incidence, as demonstrated by the far-field simulation shown in Fig. 7, where the full-vectorial FEM simulations show no detectable difference in the real experiment between s- and p-polarization states. Interestingly, the difference between the polygonal and FEM calculations at higher angles is not due to the scalar approximation; rather, it is related to the truncation of the Fourier expansion in the vertical direction that is implemented in the polygonal approach. Nevertheless, even discrepancies observed at higher angles α

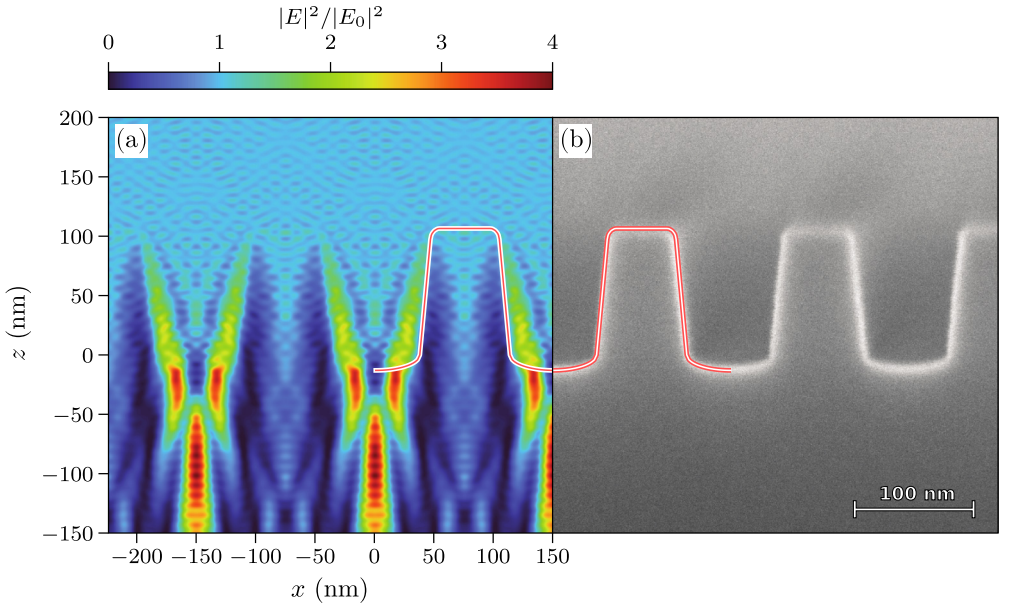


FIG. 10. (a) The near-field EED map calculated for the best-fit model, represented by the solid line, and (b) the SEM image of the silicon grating sample.

are within reasonable computational precision. Additionally, the simulations demonstrated that the polygonal approach is fully capable of nonconical diffraction simulation in the grazing-incidence geometry. Lastly, and most importantly, the polygonal approach is capable of describing the real experimental data.

IV. DISCUSSION AND OUTLOOK

With a polygonal approach, we reproduced the results of FEM and RCWA on the test model computations. Moreover, we simulated the x-ray diffraction from an actual sample, achieving a reasonable agreement with the experiment. Hence, the viability of using the proposed approach for reconstructing the parameters of nanostructures from the scattering patterns was demonstrated. This opens a compelling opportunity for applications in synchrotron nanometrology, namely, the free-form reconstruction.

In this context, the free-form approach implies the possibility of reconstructing the nanostructure's geometry without prior knowledge of its shape, nor even the knowledge to which class of shapes this structure's geometry belongs. Nowadays, the free-form approach is realized for one-dimensional structures [38] only, namely, thin films and the multilayers. The 1D free-form approach is used to reconstruct the average density profile of a nanostructure with respect to depth. Without a free-form approach, one would have to define the class of functions that describes the density gradient in depth. Then, if a wrong function were chosen, the reconstruction would be unsuccessful. The free-form approach does not rely on any analytic expression of the density profile and can handle any profile shape with a predefined resolution.

Naturally, extending the free-form approach to higher-dimensional structures is not trivial. When analyzing nanoscale gratings, one is restricted to assume a geometry model of the grating's line. Assume, for instance, one considers a simplistic square model of a line, whereas the actual line has inclined walls. Then, one modifies their model to include the sidewall angle. However, assume the actual structure is even more complex and has corner roundings. Then, one has to modify the model once again, and so on. Indeed, such a fitting process cannot be done seamlessly. At each step of increasing the reconstruction complexity, one has to devise a new model. With our polygonal approach, this process can be carried out much more easily. First, one considers a square polygon and allows the walls' position to be free parameters for the fit. If the square model is not adequate for the actual structure, one allows the position of each vertex to be fitted. Then, the tilted sidewalls automatically appear in the model. If the structure is even more complex, one should only increase the number of vertices. The only restrictions for the model are that the vertices must be ordered counterclockwise and the polygon must not cross itself.

An obstacle to realizing a free-form analysis based on the polygonal approach is the computation speed. The conventional RCWA approach is based on a one-dimensional Fourier transform of the scattering potential, while the proposed approach assumes a two-dimensional structure. This leads to the eigenproblem of a larger size [cf. Eq. (8)] and necessitates solving a QEP. The computational complexity problem can be circumvented by developing an algorithm to find fast approximate solutions to the eigenvalue problems, which takes into account the matrix structure of the QEP. Indeed, the information entropy of the characteristic matrix in our case is reduced compared with an arbitrary matrix of the same size, as the arising matrix $C_{[\chi]}$ is Toeplitz (has entries constant along diagonals). It hints at the possibility of devising a numerically effective algorithm specific to our type of QEP. We can restate that in terms of physical intuition. The farther the wave modes are separated in the reciprocal space (the difference between diffraction order numbers), the less these modes interfere. Thus, there is no need to take into account the interference

between each mode. Hence, the approximate solution should exist.

If there is an approximate solution, parity between decay and gain modes may no longer be guaranteed, as in the cases described in Refs. [39,40]. In this case, the asymmetric BDC solution [Eq. (16)] could become useful. We focus our future research on finding a numerically efficient solution and realizing a free-form approach for reconstructing the grating's structure.

V. CONCLUSION

We have described a Fourier modal approach to the simulation of scattering in lamellar gratings. It is based on a combination of ideas from rigorous coupled wave analysis and dynamical diffraction theory. This allowed us to consider the grating as it is, without approximating it with layered models and without constraining the many beam interactions. We also implemented polygons to model the grating profile. This, in turn, allows us to easily calculate scattering on gratings of arbitrary complexity. This is a substantial improvement over a conventional approach in which the profile geometry model is fixed and any new shape requires a new model to be devised. We have compared the proposed approach with rigorous coupled wave analysis and the finite element method. All considered approaches produce solutions of similar order of accuracy with relative L_2 norms less than 2% compared with the finite-element solution. Thus, we conclude that the proposed approach is suitable for quantitative simulations. Furthermore, we have compared the simulations performed by the proposed approach with the synchrotron scattering experiments. The simulations are in qualitative agreement with the experiment. The proposed approach can be employed as a nanometrology tool for future microelectronics and in other fields of science and technology. This will require further research to improve the computational speed. However, the present study takes the first steps in this direction. Namely, the generalized solution of the boundary-condition problem has been found, which may allow the implementation of faster algorithms in the future.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available because of legal restrictions preventing unrestricted public distribution. The data are available from the authors upon reasonable request.

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